

4 Critical path analysis	4.1	Modelling of a project by an activity network, from a precedence table.	Activity on arc will be used. The use of dummies is included.
	4.2	Completion of the precedence table for a given activity network.	In a precedence network, precedence tables will only show immediate predecessors.
	4.3	Algorithm for finding the critical path. Earliest and latest event times. Earliest and latest start and finish times for activities. Identification of critical activities and critical path(s).	Calculating the lower bound for the number of workers required to complete the project in the shortest possible time is required.
	4.4	Calculation of the total float of an activity. Construction of Gantt (cascade) charts.	Each activity will require only one worker.
	4.5	Construct resource histograms (including resource levelling) based on the number of workers required to complete each activity.	The number of workers required to complete each activity of a project will be given and the number of workers required may not necessarily be one.

Topics	What students need to learn:		
	Content		Guidance
4 Critical path analysis <i>continued</i>	4.6	Scheduling the activities using the least number of workers required to complete the project.	